

Benchmark Workflow List

We incorporate ten representative workflows collected from two widely used Scientific Workflow Systems: Galaxy and Nextflow. The selected Galaxy workflows cover common bioinformatics analysis tasks such as SNP analysis, peak-to-gene mapping, gene overlap analysis, raw data quality control, and bacterial genome annotation. The Nextflow set includes five nf-core workflows representing practical sequencing and genomics use cases, including quality control, metadata retrieval, methylation sequencing, RNA-seq analysis, and phylogenetic placement. Together, these workflows provide diverse examples for evaluating workflow generation across different systems and application scenarios.

Table: Overview of the used workflows.

SWS	WID.	Workflow	Link
Galaxy	W1	Which coding exon has the highest number of single-nucleotide polymorphisms on human chromosome 22?	[1]
	W2	How to get from peak regions to a list of gene names?	[2]
	W3	Do genes on opposite strands ever overlap? If so, how often?	[3]
	W4	How can the quality of NGS raw data be assessed, what parameters should be checked, and how can the quality of the dataset be improved?	[4]
	W5	Which genes are on a draft bacterial genome, and which other genomic components can be found on a draft bacterial genome?	[5]
Nextflow	W1	nf-core/demo: This bioinformatics pipeline performs quality control and reporting on sequencing data.	[6]
	W2	nf-core/fetchngs: It is a bioinformatics pipeline to fetch metadata and raw FastQ files from databases.	[7]
	W3	nf-core/methylseq: This workflow is used for Methylation (Bisulfite) sequencing data.	[8]
	W4	nf-core/rnaseq: This workflow is used to analyse RNA sequencing data with a reference genome and annotation.	[9]
	W5	nf-core/phyloplace: This workflow performs phylogenetic placement with EPA-NG.	[10]

References

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